

The screenshot couldn't be generated

Hello! How is it going?



not bad kinda fine

good

for zandalar

goodness

more coffee

Agenda

- Recall last session
- Discuss current topic
- Practice
- Quiz

Total Recall



What do you remember most from few previous workshops? (short answer)

Junit

unit test, mokito

mock object

build

Types of Testing:

- Unit testing
- System testing
- Integration testing
- Acceptance testing
- Performance testing
- Regression testing
- Security testing
- Load testing
- End-to-end testing
- And more



Next topic

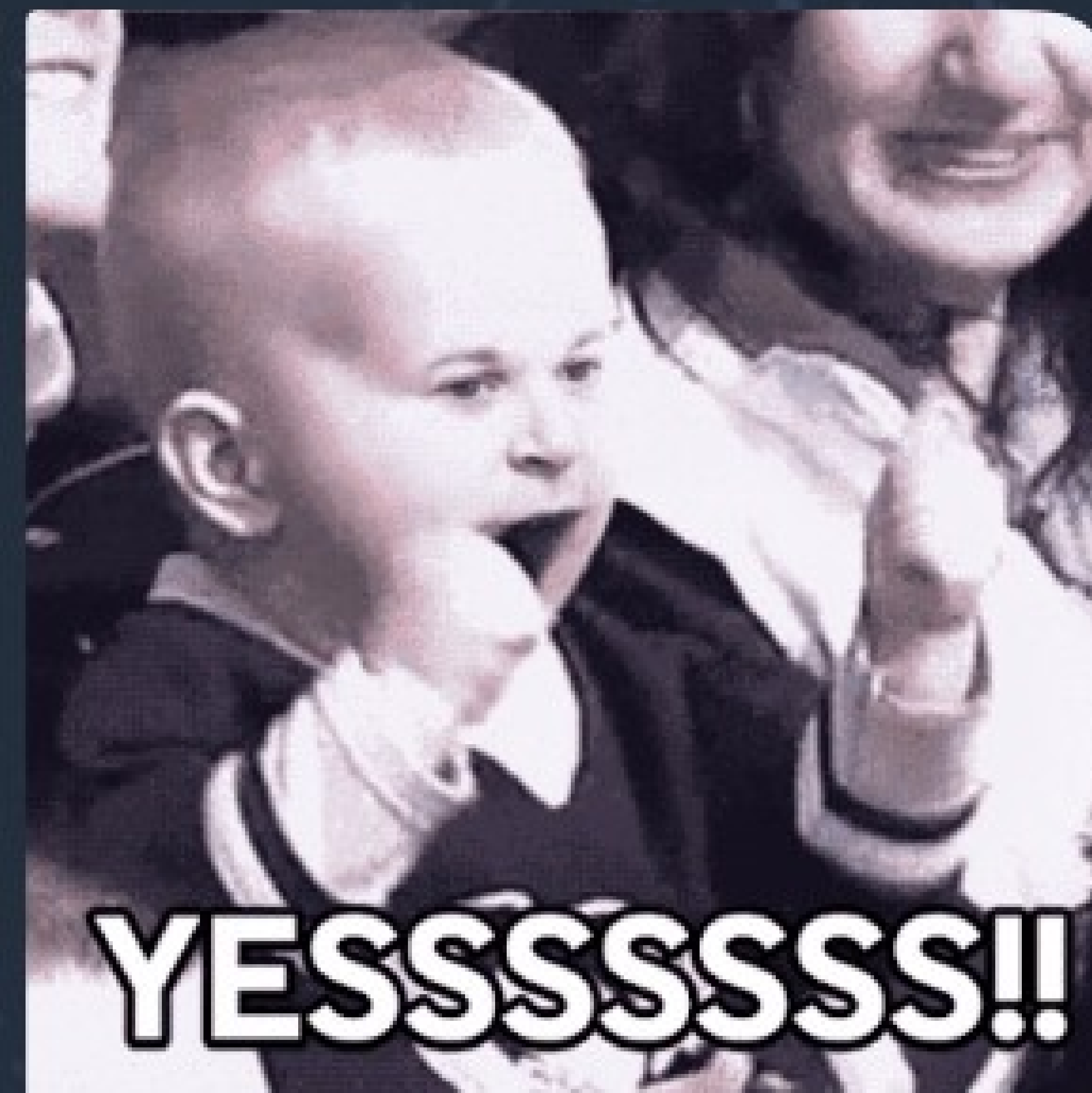


Random fact

Giraffes are 30 times more likely to get hit by lightning than people

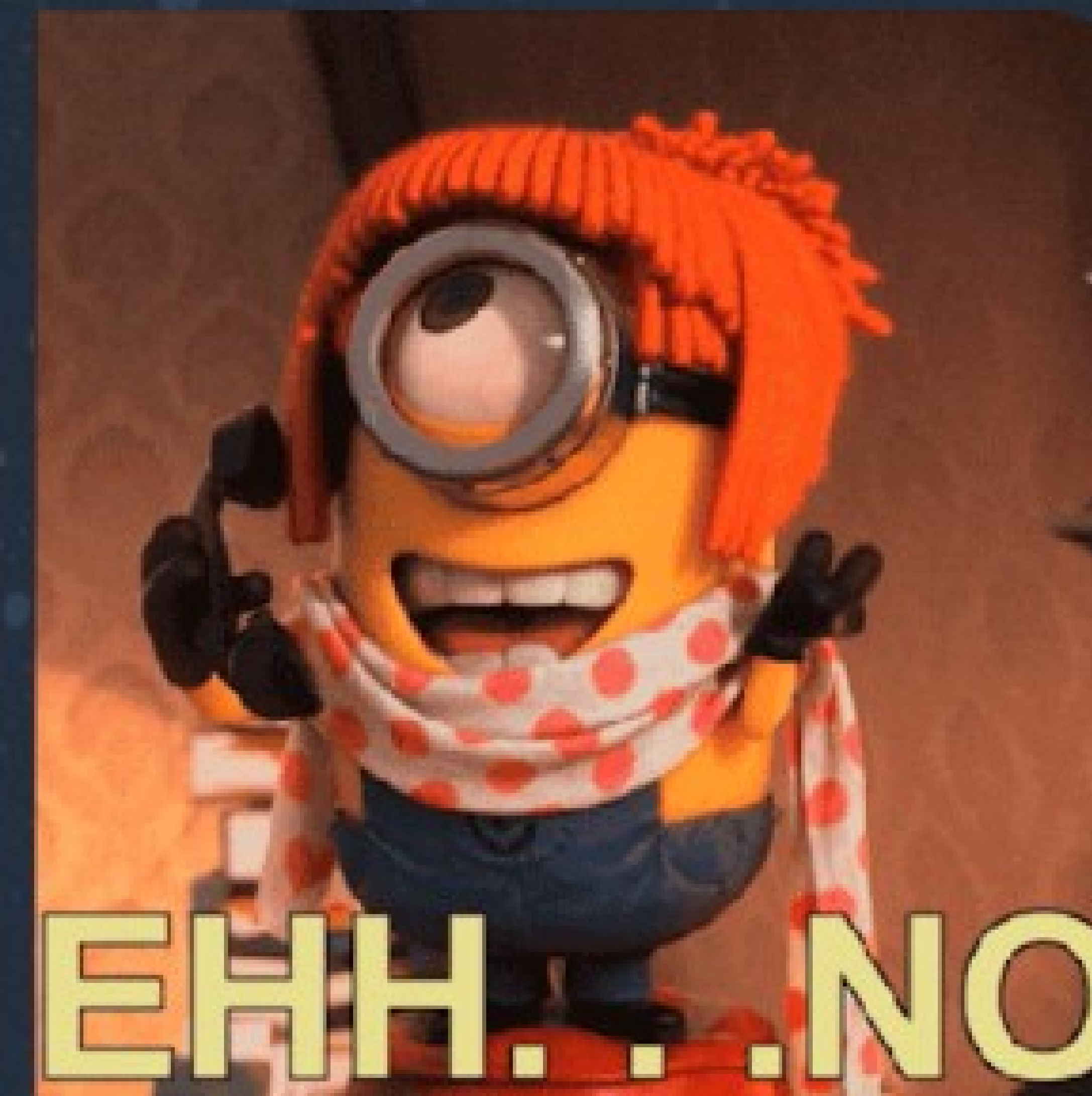
Have you looked through the materials?

6



yes

2



no

Writing Basic Tests With JUnit

ARRANGE -> ACT -> ASSERT

Annotations:

- AAA
- @Test
- @DisplayName("Some name")

Assertions

- assertEquals(expected, actual) && assertNotEquals(expected, actual)
- assertSame(expected, actual)
- assertNull(actual) && assertNotNull(actual)
- assertTrue(actual) && assertFalse(actual)
- assertThrows(Throwable.class, () -> testingObject.testingMethod())
- assertAll(() -> testingObject.testingMethod_1(), () -> testingObject.testingMethod_2())
- assertSome(() -> testingObject.testingMethod_1(), () -> testingObject.testingMethod_2())
- assertTimeout(Duration.ofNanos(1000), () -> testingObject.testingMethod())
- assertEqualsArrays(expectedArray, actualArray)
- assertEquals(delta, failMessage)

Assumptions:

- `assumeTrue(assumption)`
- `assumeFalse(assumption)`
- `assumingThat(assumption, () -> assertTrue(actual))`
-

Assumption

- Assumption met: test runs
- Assumption not met: test aborted

Assertion

- Assertion true: test succeeds
- Assertion false: test fails

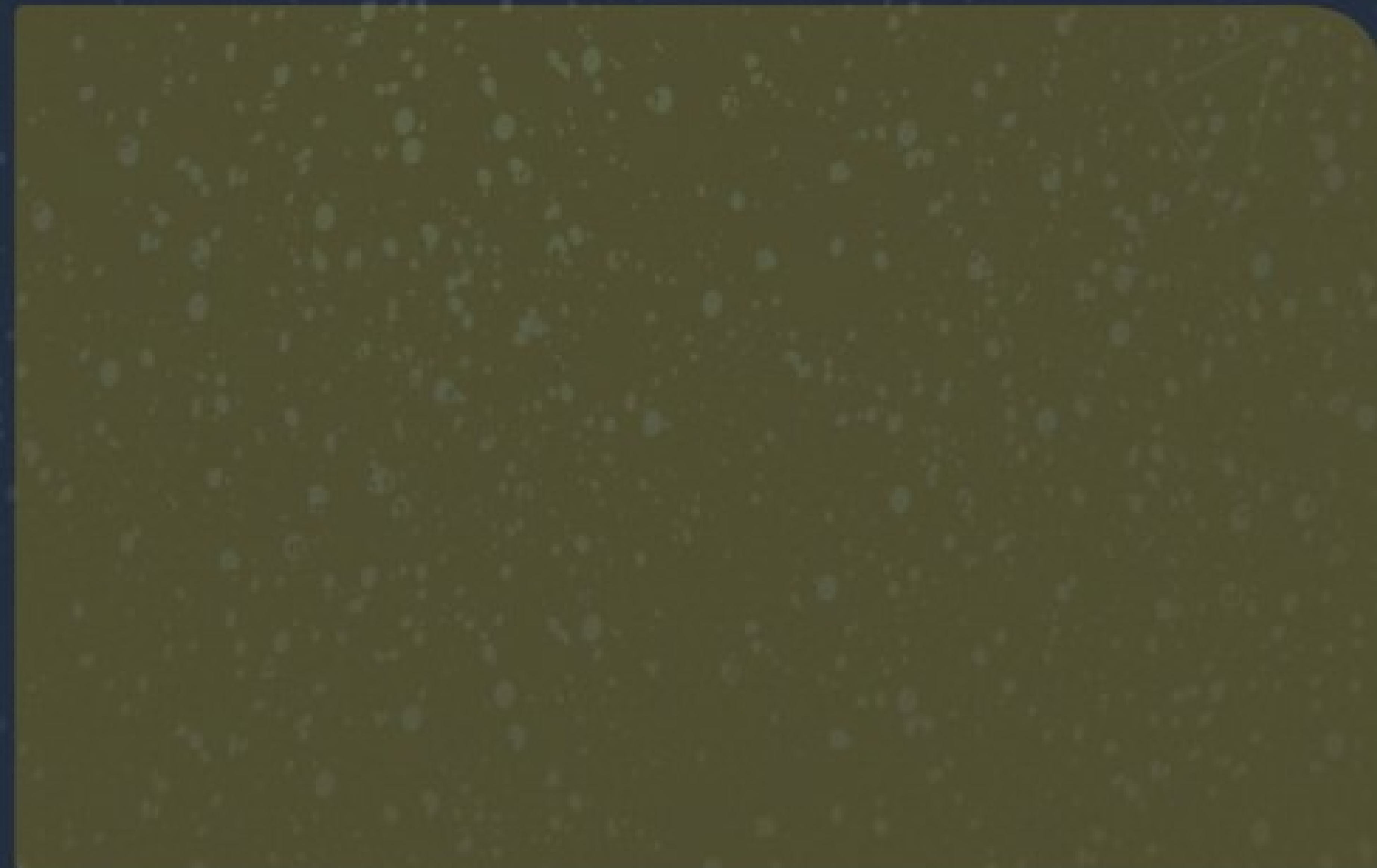
What happens when an assumption fails?

8 ✓



The test will be aborted.

5 ✗



The test will be ignored.

0 ✗

The test will fail.

Test Order

- Without specifying, we cannot predict the order
- Usually, this is not a problem: tests should be independent
- If you do need to order your tests, JUnit 5 comes with an elegant solution
- How?

Use annotations

- `@TestMethodOrder(MethodOrderer.OrderAnnotation.class)`
- `@Order(1)`
- `@Order(2)`
- `@Order(..)`

Nested test class

use @Nested above nested test class

More Advanced Uses of JUnit

Code JUnit 5 examples:

- Dependency injection
- @RepeatedTest
- @ParametrizedTest
- @ValueSource
- @EnumSource
- @CsvSource
- @CsvFileSource

JUnit 4:

- @RunWith(JUnitParamsRunner.class)
- @Parameters(method = "testValues")
- @Parameters({ "1, 1, 2", "5, 5, 10", "10, 15, 25", "100, 200, 300" })

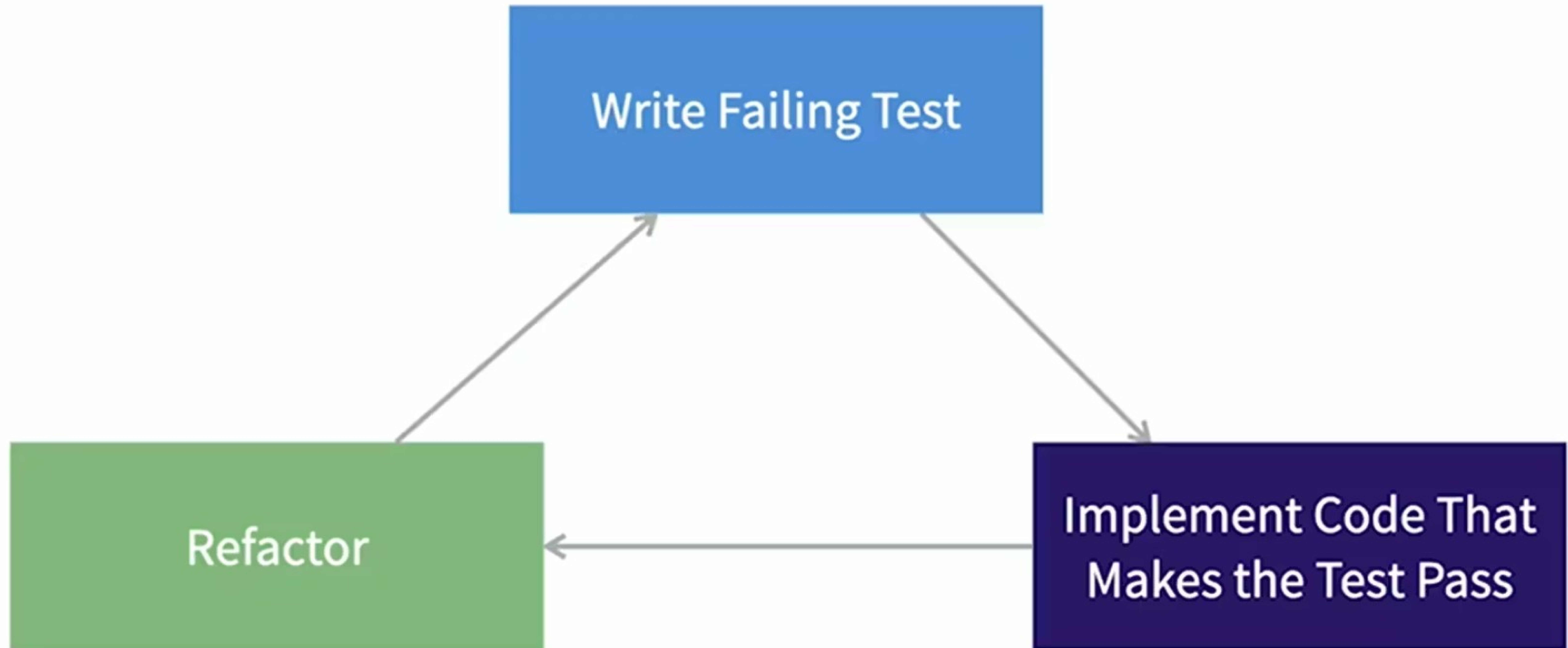
More:

- Timeout tests - [CalculatorImplTimeoutTest.java](#)
- Parallel Execution - [CalculatorImplParallelExecutionTest.java](#)
- @BeforeAll , @AfterAll , @BeforeEach - [CalculatorImplTest.java](#)
- SureFire: mvn surefire-report:report - [pom.xml](#)
- test on condition: mac/win/linux/jdk8/jdk11/... - [CalculatorImplConditionTest.java](#)
- @Disabled("Flacky test. Require fix.")

Testable Code

- Test small units: split up code
- Loosely coupled
- Single responsibility principle
- High cohesion

Test-Driven Development (TDD)

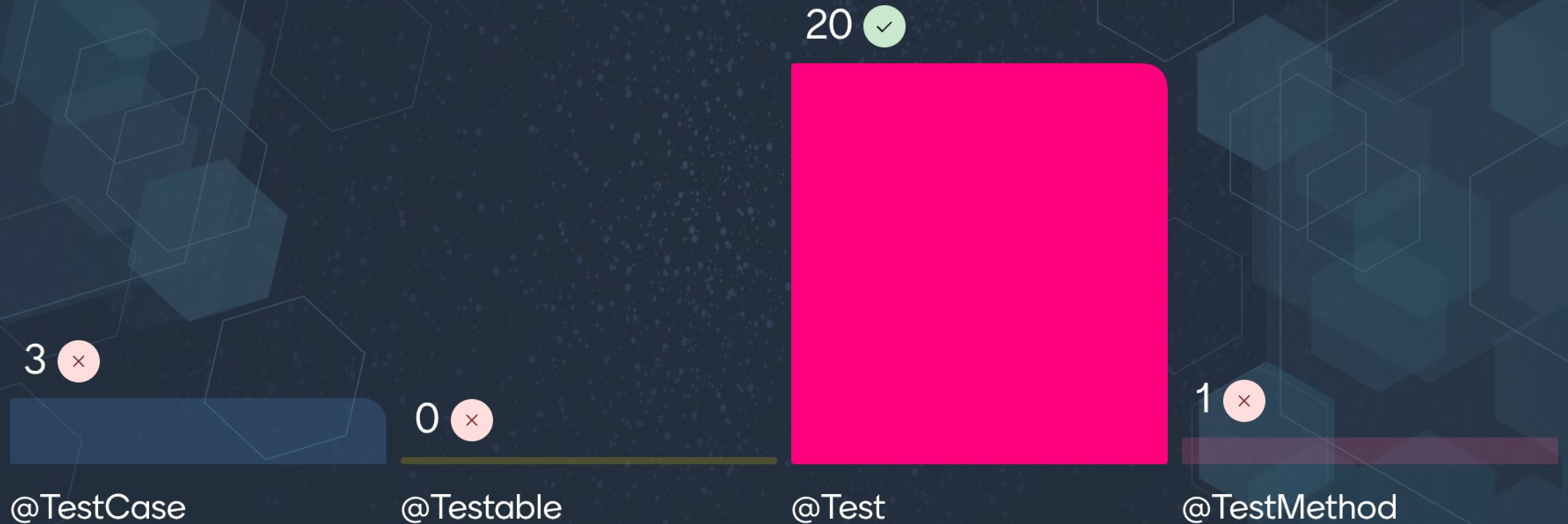




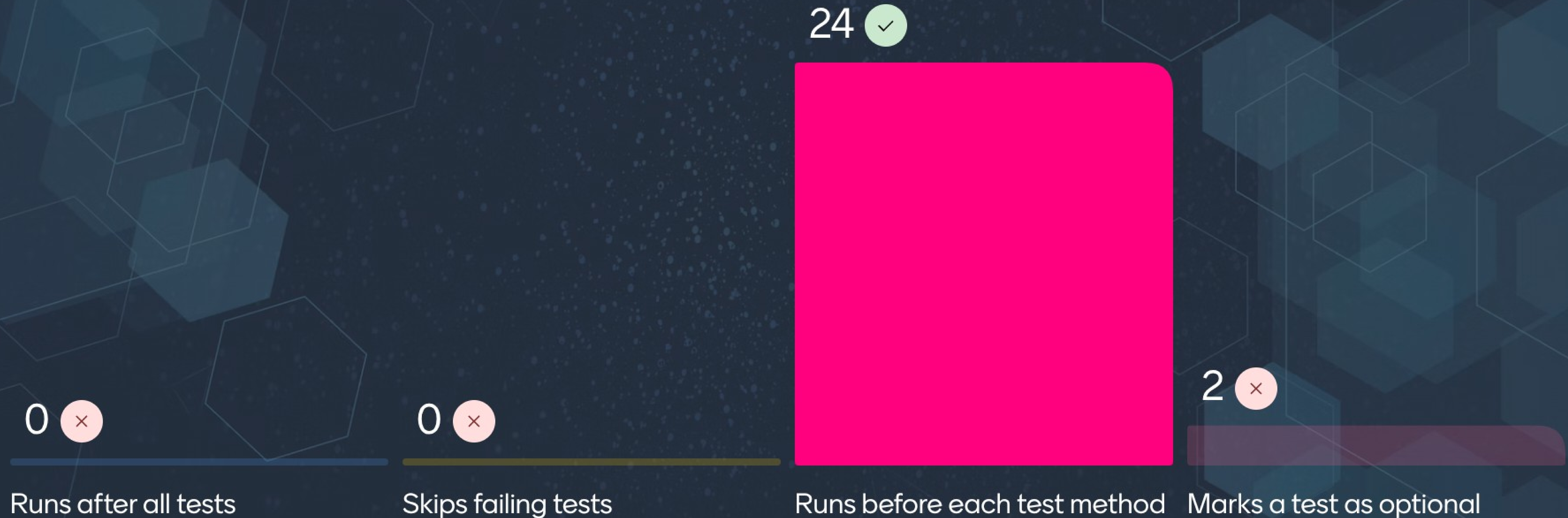
Quiz time

QUIZ

How is a test method annotated in JUnit5?



What does @BeforeEach do in JUnit5?



What is the purpose of @BeforeAll?

23 ✓



0 ✗

Executes once after tests

0 ✗

Verifies individual test cases

0 ✗

Logs the test execution time

What annotation enables parameterized tests in JUnit5?

0 ✗

@TestValues

1 ✗

@TestSubject

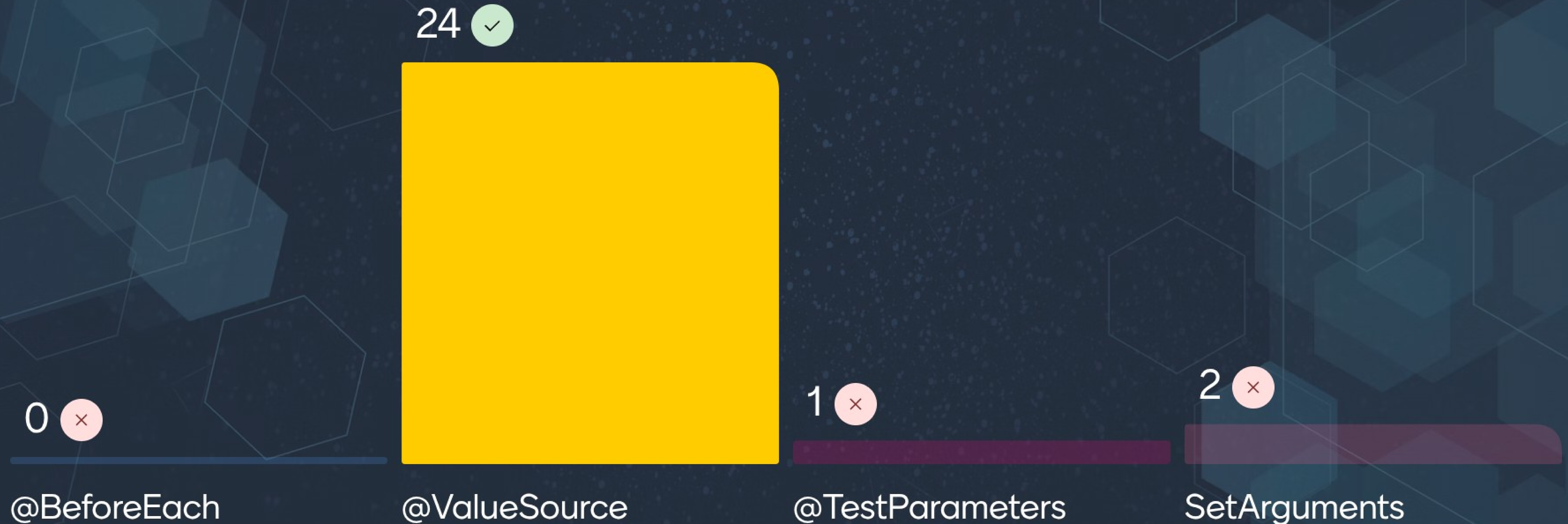
1 ✗

@ScriptTest

25 ✓

@ParameterizedTest

Which annotation provides arguments for a parameterized test?



Which JUnit method verifies an expected exception?

1 ✗

assertNotNull()

2 ✗

afterEach()

23 ✓

assertThrows()

2 ✗

assertContains()

What does Assertions.assertEquals() verify?

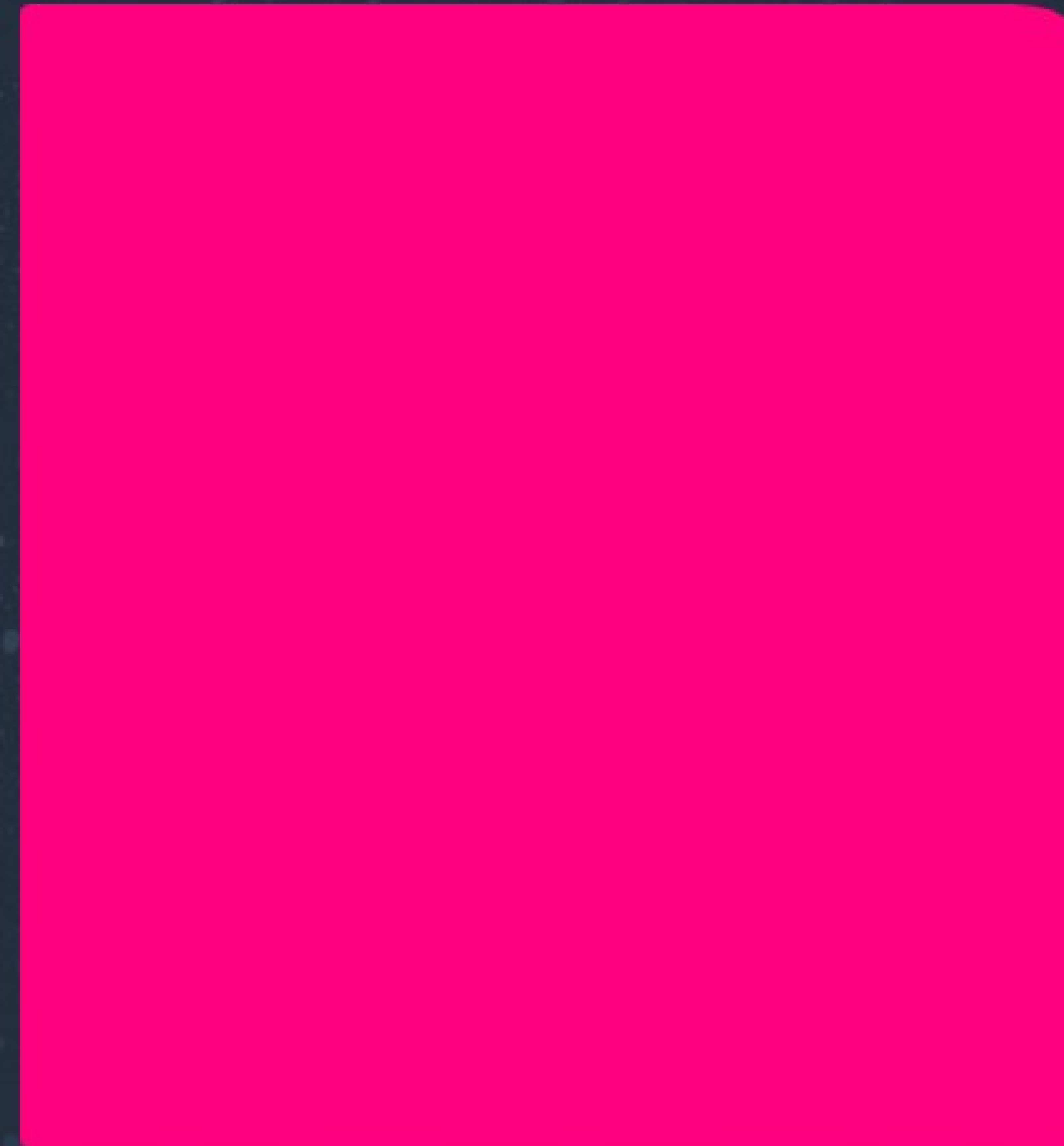
27 ✓

0 ✗

Exception throwing

0 ✗

Variable nullity



0 ✗

Code performance

Expected and actual values are
equal

What is key in test-driven development (TDD)?

8 ✓



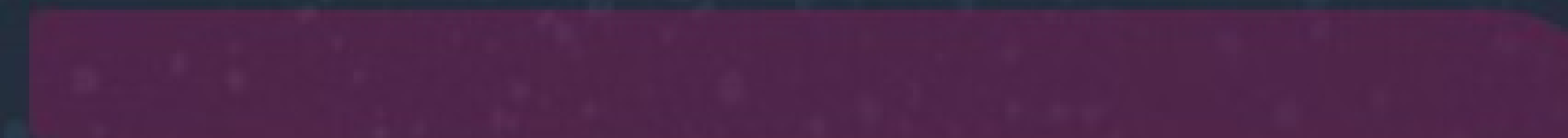
Write failing tests and then
implement logic

16 ✗



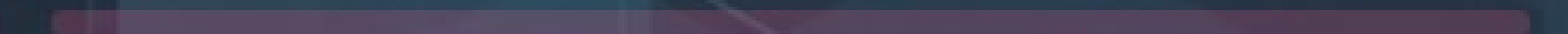
First write tests, then code

1 ✗



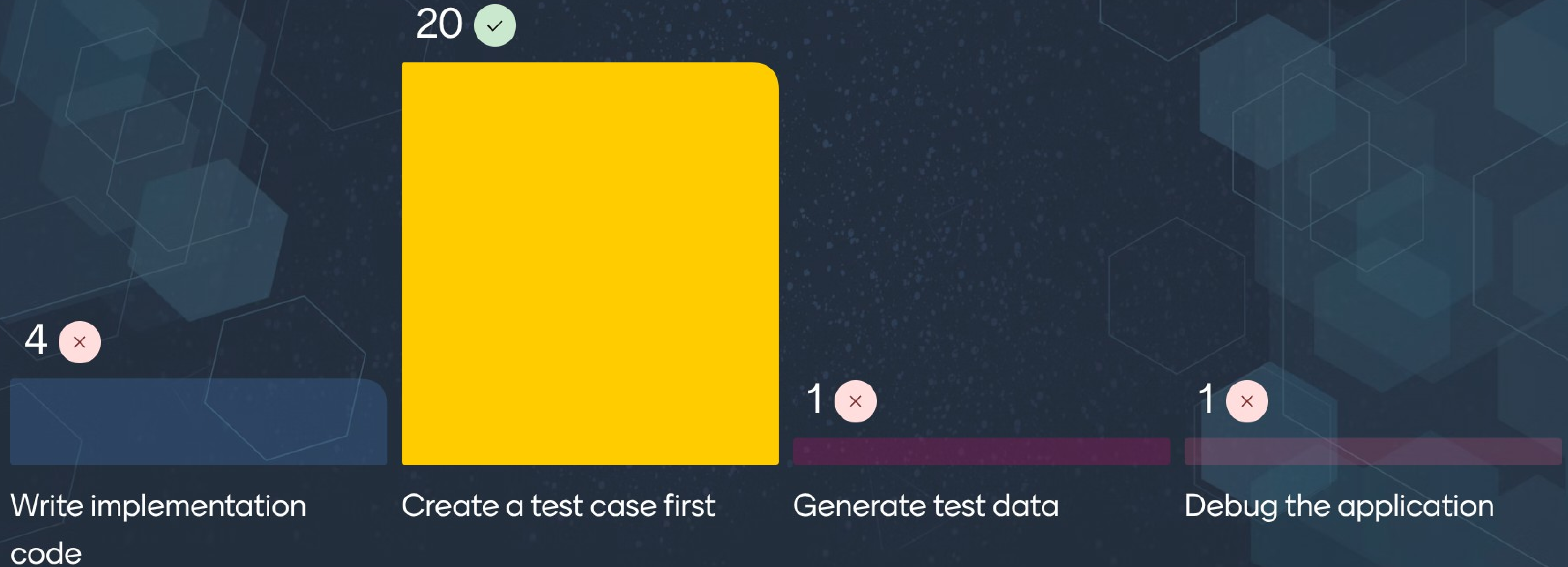
Write all tests at once

0 ✗



Run tests after deployment

Example: What does TDD workflow start with?



What is the purpose of @DisplayName in JUnit5?

24 ✓



1 ✗

Control execution time

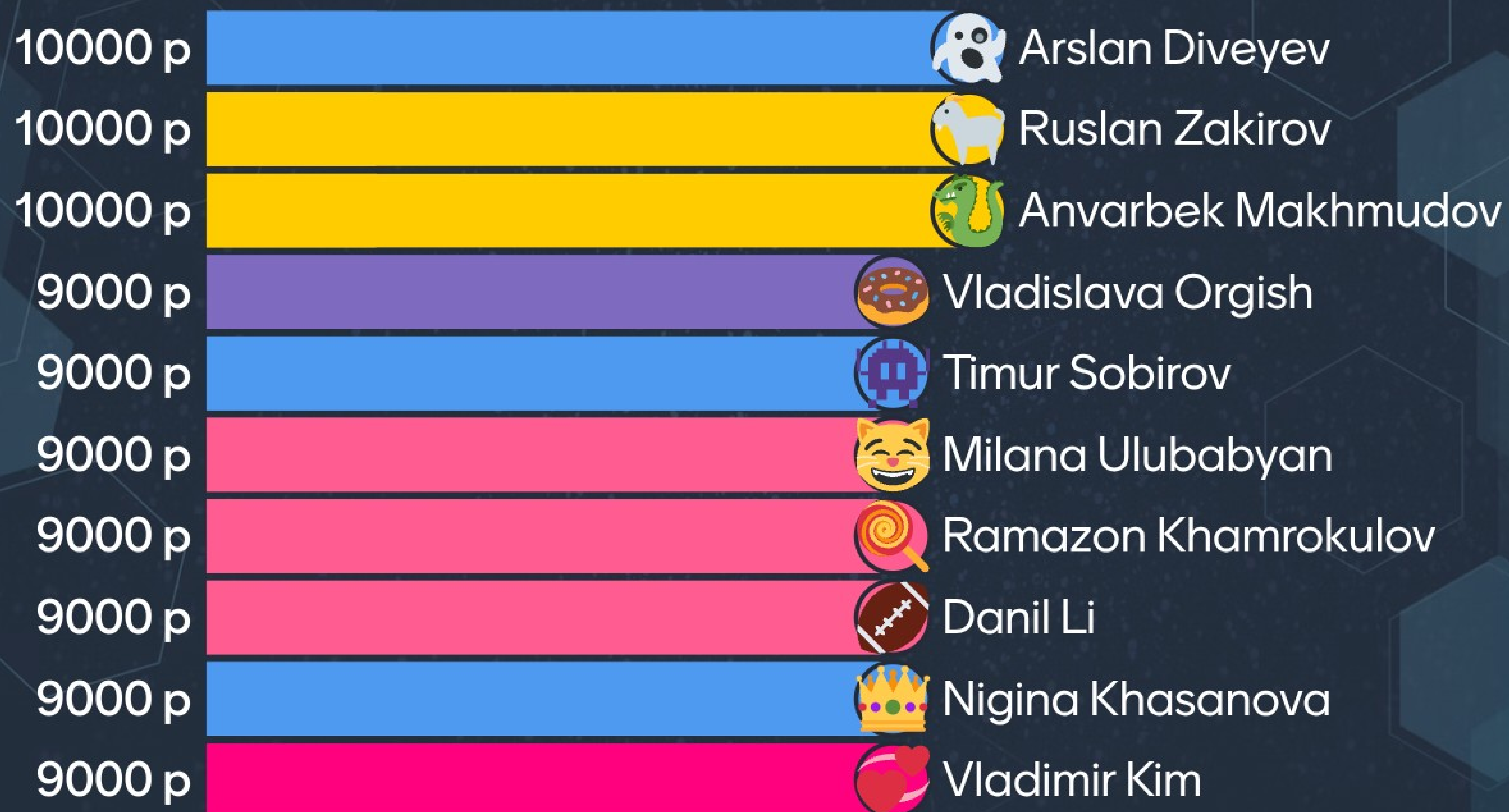
0 ✗

Validate test data

1 ✗

Import symbols automatically

Quiz leaderboard



Q&A part

0 questions
0 upvotes

